

**Curing Cancer in a FLASH:
A Finite-Element Computational Analysis of
an Ultrasonic Detection Array Setup for
Ultra-High Dose Rate (FLASH) Measurements
and Absolute Dosimetry**

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Abstract

FLASH Radiotherapy is a promising radiation therapy modality, delivering radiation dose at an ultra-high dose rate (≥ 40 Gy/s) compared to Conventional Radiotherapy (0.01 – 0.40 Gy/s) while minimizing irradiation damage to healthy tissue surrounding a tumor. However, existing radiation dose imaging technology using thermistors cannot determine, with significant precision and millisecond-regime time delays, whether a surface dose achieved FLASH-level dose rates. A proposed solution involves propagating an ultrasonic wave orthogonal to a radiative beam from an electron linear accelerator in a water phantom to observe potential phase modulation of the ultrasound, which can be de-modulated to determine the delivered dose rate. This investigation aims to conduct a Finite-Element Method (FEM) analysis of the ultrasonic detection array setup proposed via simulations of a complete thermoacoustic model on the COMSOL® Multiphysics 6.1 software to confirm phase modulation of the ultrasonic signal. First, a simplified water phantom was constructed, bounded by physical and material parameters and surrounded by a Perfectly Matched Layer. Then, a radiative beam was devised by applying multipulse laser machining principles to a heat flux condition. Similarly, the ultrasonic signal was adapted from a High-Intensity Focused Ultrasound (HIFU) pulse, applying diagnostic ultrasound principles to switch from HIFU's use case as a therapeutic modality. The Pressure Acoustics and Heat Transfer modules were accordingly coupled to obtain temperature change, acoustic pressure, and phase of the ultrasonic wave upon simultaneously propagating the radiative beam and ultrasonic signal. The simulation results show a significant phase modulation of the ultrasonic signal after reaching the radiative beam. Upon interaction with the heated region (10 mK temperature difference achieved from a 40 Gy/s beam), the ultrasonic signal initially faced an increase in frequency via compression before decreasing below the original frequency. A decrease in the number of peaks was additionally observed, indicating phase modulation. Moreover, slight amplitude modulation in the heated region was observed in the transient domain. While the presence of thermoacoustic shocks may be attributed to FEM meshing, this behavior is validated by existing literature. The computational model results are also validated by preliminary results observed in experiment with close agreement. The results of this investigation may be applied in methods for Absolute Dosimetry of surface doses, Holographic Interferometry of radiotherapy beams, radiation calorimetry, and high-precision FLASH measurements to make FLASH Radiotherapy a clinical modality.

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Introduction

Subject

The effect of Ultra-High Dose Rate (UHDR) Radiation was first discovered in 1959 by D. Dewey and J. Boag, when Salmonella bacteria treated with increasing rates of X-ray radiation dose experienced a decrease in radiation sensitivity and an increase in survival rate. This led to the formal definition of FLASH Radiotherapy (FLASH-RT) as a modality in 2014, after in vivo experiments on mice with UHDR Radiation [1]. In 2019, FLASH-RT was tested on its first patient, a 75-year-old with multi-resistant CD30+T-cell cutaneous lymphoma. The patient had been treated 110 times with localized CONV-RT, which failed to completely treat their cancer; one session of FLASH-RT led to complete tumor response in 36 days after irradiation, and their cancer was successfully in remission for next 5 months [2]. FLASH-RT has further shown preclinical success treating sarcoma and skin cancers [3], recurrent tumors in head and neck cancers [4], symptomatic bone metastases [5, 6], and deep-seated tumors [7, 8]. As such, FLASH-RT has begun to create a paradigm shift in radiation oncology practices globally.

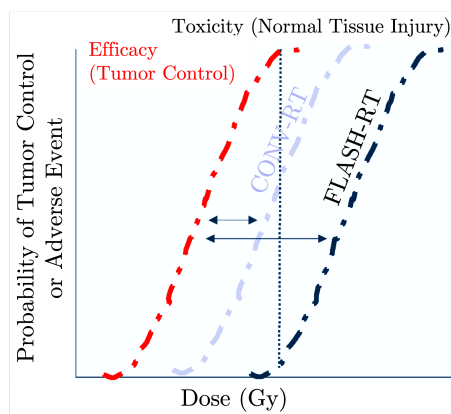


Figure 1: Therapeutic Window for FLASH-RT and CONV-RT [3]

From “Flash Radiotherapy: What Can FLASH’s Ultra High Dose Rate Offer to the Treatment of Patients with Sarcoma?,” by K. A. Cengel, M. M. Kim, E. S. Diffenderfer, & T. M. Busch, 2024, *Seminars in Radiation Oncology*, 34(2), p. 219. <https://doi.org/10.1016/j.semradonc.2024.02.001>

FLASH Radiotherapy (FLASH-RT) delivers radiation dose at an ultra-high dose rate (≥ 40 Gy/s in under 200 ms) in relatively few fractions [9, 10], compared to Conventional Radiotherapy (CONV-RT), delivering dose at 0.01 – 0.40 Gy/s dose rates for over a minute in a high number of fractions [1]. FLASH-RT minimizes irradiation damage to healthy tissue around a tumor unlike CONV-RT [11], as seen in Figure 1. Though FLASH-RT’s exact mechanism for this effect is unknown, several theories have been proposed. One class of theories, the physicochemical processes, notably includes oxygen depletion and reactive oxygen species [12].

The former theory suggests that FLASH-RT causes the rapid consumption of local oxygen upon irradiation, causing the cell to enter transient radiation-induced hypoxia. As a result, free radicals created by radiation cannot cause permanent damage to the cell, minimizing damage to healthy tissue [13].

The latter theory has become more promising in recent years. Cells that normally undergo Glycolysis have a hyper-efficient pathway known as Oxidative Phosphorylation (OXPHOS), which consumes toxic Reactive Oxygen Species (ROS) that are left in the cell cytoplasm and can cause cellular destruction when left undisturbed. Both pathways are necessary for a cell to produce energy in the form of Adenosine Triphosphate (ATP). CONV-RT is found to uniformly disrupt OXPHOS, increasing toxic ROS and upregulating Glycolysis in tumor cells, which can temporarily promote the proliferation of cancer cells while causing damage to healthy cells around a tumor [14]. CONV-RT eventually results in normal tissue toxicity, leading tumors to die through a lack of energy production. In contrast, FLASH-RT is thought to minimize OXPHOS activity in healthy cells, lowering both ATP and ROS levels. This state is known as metabolic quiescence, which prevents tissue toxicity effects in non-cancerous tissue [11].

Another class of theories, the biological processes, notably includes the immune hypothesis [15]. CONV-RT has been found to cause 90-100% death to circulating immune cells, whereas FLASH-RT causes only 5-10% of death to circulating immune cells, suggesting that FLASH-RT’s minimization of irradiation damage is linked to the sparing of immune cells [16]. Other studies have conducted investigations on the short timescale of FLASH-RT irradiation directed at a tumor, which could also cause strong localization of irradiation damage towards tumor cells [17].

Purpose

Verifying dose rate within a volume for FLASH-RT is critical for clinical adoption, but not possible with current dosimetry hardware or measurement standards. This work seeks to address both shortcomings by developing an absolute dose and dose-rate detection principle based on ultrasound. Specifically, by conducting a Finite-Element Method (FEM) analysis of a proposed detection setup via simulations of a complete thermoacoustic model on the COMSOL® Multiphysics 6.1 software to confirm phase modulation of an ultrasonic signal.

Scope

The success of this research investigation will allow for the precise observation of delivered radiation dose and dose rate, which will help elevate FLASH-RT's standard to a therapeutic modality by providing a patient/insurance guarantee that FLASH-RT was successfully completed. Furthermore, this study will provide a method to investigate the cellular causes for the FLASH biological mechanism through ultrasonic imaging. As a whole, this investigation will improve both scientific and commercial understanding of Absolute Dosimetry of surface doses, Holographic Interferometry of radiotherapy beams, radiation calorimetry, and high-precision FLASH measurements.

Development

The methods followed throughout this investigation will be presented first to provide the reader with a foundation on creating simulations through the COMSOL® Multiphysics 6.1 software. Any assumptions made that may impact the study will be discussed initially. The general methodology followed will be listed before the results of this study to guide the reader through the research investigation. An analysis of the simulations conducted will be discussed to justify the success of the complete Thermoacoustic model, and conclusions will be drawn. References will be shown to enhance the reader's comprehension on the subject matter.

Methodology

Methods

All simulations in this study were conducted using the COMSOL® Multiphysics 6.1 software. For consistency, the metric system was followed in all measurement quantities.

In all simulations, a water phantom was created with dimensions $2\text{ cm} \times 2\text{ cm} \times 2\text{ cm}$, modeled to fit inside a Perfectly Matched Layer (PML) plexiglass water bath. A Radiative Beam and Heat Flux condition was placed in a rectangular cross section of diameter 6 mm and height 1.6 cm, constructed with similar computational techniques seen in multi-pulse lasers [18]. The speed of sound in water was redefined to follow Chen and Millero’s UNESCO speed of sound equation. Furthermore, the Ultrasound pulse propagated was modified from a High-Intensity Focused Ultrasound (HIFU), changing its use-case from a therapeutic modality to a diagnostic modality.

Design models used for testing functionality of the Acoustic module alone had an air ambient of same dimensions as the water phantom. In these models, Ultrasound was propagated from the air ambient before entering the water phantom. Models used for testing functionality of the Heat Transfer (Thermal) module eliminated the air ambient altogether. All external boundary conditions were Perfectly Matched Layers, to prevent confounding of results from wave reflections.

A centered cut-line oriented horizontally was taken to measure the Acoustic Pressure of an Ultrasonic pulse propagating in the x -direction, since Acoustic Pressure is directly proportional to a wave’s propagation in spatial coordinates and hence would showcase similar effects of modulation (Phase, Frequency, and Amplitude). Another cut-line was placed vertically, inside the Radiative Beam and Heat Flux condition to maintain the milli-Kelvin regime heating of a 40 Gy/s pulse necessary for FLASH-RT.

Procedures

Initially, existing literature was reviewed to gain insight in current experimental methods involved with FLASH-RT, background in relevant oncology studies and simulation tactics, and ideas on creating a detection scheme for precise measurement of delivered dose and dose-rates to a cancerous cell. Upon completion, the COMSOL® Multiphysics 5.6 Software was explored to identify requirements and methods needed to create a complete Thermoacoustic model. To that end, the methodology behind creating plots and displaying relevant data was learned and eventually applied

to carry out this research study. Finally, the following technical report was written to convey all results obtained during the duration of this study in a coherent manner.

Results

It is known that the speed of sound in water increases with temperature roughly linearly, as seen through a graph of Chen and Millero's UNESCO speed of sound equation in Figure 2:

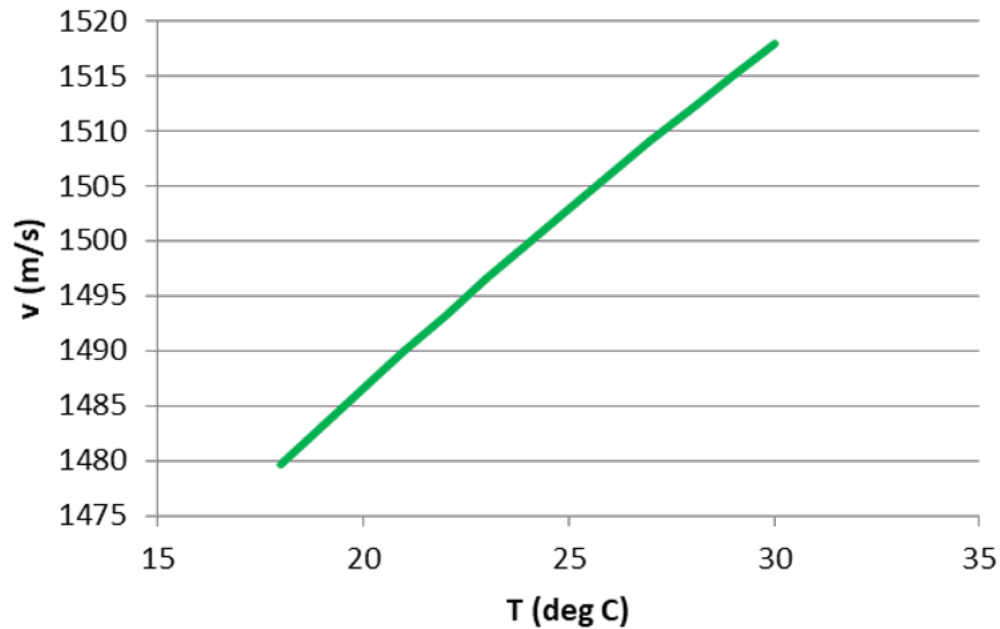


Figure 2: Speed of Sound in Water vs. Temperature

“Ultrasound for Calorimetry, Figure 1.” Graphic created by Dr. Ronald E. Tosh (Research Advisor), National Institute of Standards and Technology, 2025.

As such, when heating occurs in a water phantom, the speed of sound will also increase. Thus, if we inject a 5 MHz carrier into a water phantom and apply radiation heating, we should expect to see a time-dependent phase shift in the Ultrasonic signal. Furthermore, periodic application of radiation will frequency-modulate the carrier wave, as shown in Figure 3.

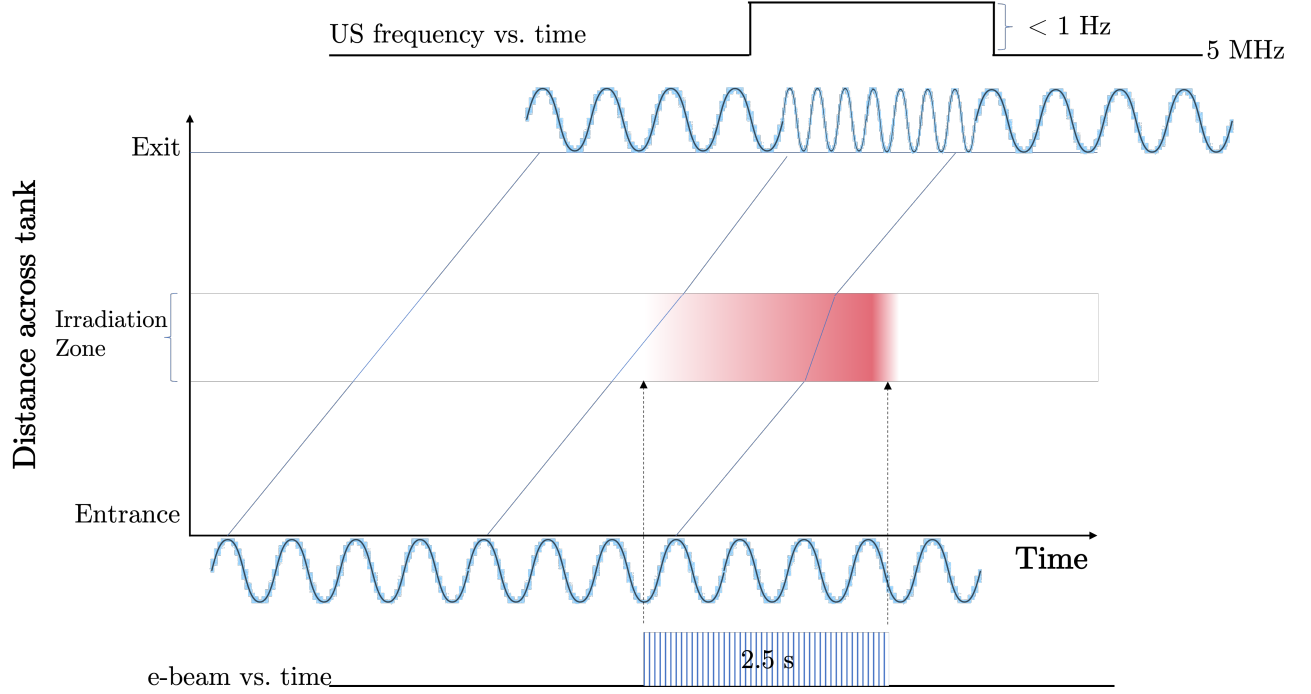


Figure 3: Ultrasonic Frequency vs. Time across Undisturbed and Irradiated Regions

“Ultrasonic Calorimetry for Dose to Water in FLASH Electron Beams, Figure 2.” Graphic created by Dr. Ronald E. Tosh (Research Advisor), National Institute of Standards and Technology, 2024.

We hence propose the following detection principle: to propagate an ultrasonic wave orthogonal to a radiative beam in a water phantom to observe potential phase modulation of the ultrasound for de-modulation, determining the delivered dose rate. Note that a water phantom is used to replace actual tumor cells, as it is computationally- and experimentally-expensive to construct human tumor cells, and human soft tissue is mostly (generally above 70%) water. To meet experimental standards, the ultrasonic carrier will be a 5 MHz wave propagated from a piezoelectric transducer. We propose this approach from an experimental precision standpoint. From Figure 2:

$$v(T) \approx 1394 + 4.5T \Rightarrow \Delta T \approx 0.2391 \frac{\text{mK}}{\text{Gy}}$$

Hence, a 40 Gy/s electron beam would cause an approximate heating of 9.564 mK in the water phantom, yielding a slight phase modulation of the carrier wave. Traditional methods for analyzing delivered dose would involve observing the change in the Time of Flight of the signal [19, 20]:

$$TOF \approx 7 \frac{\mu\text{s}}{\text{cm}} \Rightarrow \Delta TOF \approx 10 \frac{\text{ps}}{\text{Gy}}$$

A change in the picosecond regime is relatively difficult to observe experimentally, and requires specialized equipment. To bypass this, instead of observing Time of Flight for dose and dose-rate,

we can look to phase:

$$\Delta\phi = \Delta(\omega \cdot TOF) \approx \omega \cdot \Delta TOF \approx 18 \frac{\text{mdeg}}{\text{Gy}}$$

A change in the millidegree regime is relatively easier to measure than a change in the picosecond regime via a lock-in amplifier instrument. This forms the basis of the following conceptual scheme to be simulated and analyzed thermoacoustically:

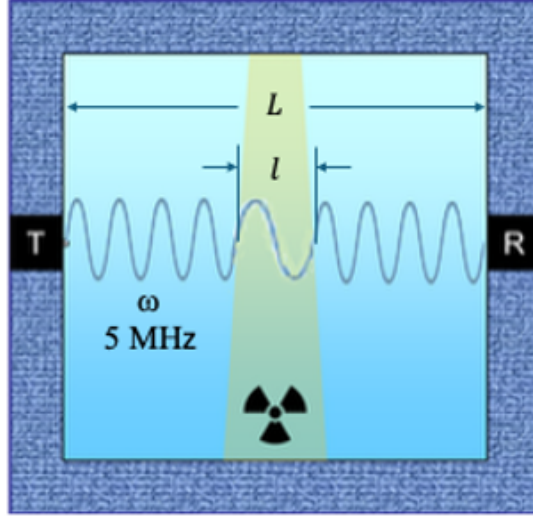


Figure 4: Proposed Setup

“Ultrasound for Calorimetry, Figure 2.” Graphic created by Dr. Ronald E. Tosh (Research Advisor), National Institute of Standards and Technology, 2025.

Thermal Model Validation

Before constructing the final model, we can individually construct the thermal and acoustics portions. Figure 5 below shows a Temperature surface plot of our water phantom, with a vertical radiative beam modeled by a Heat Flux condition coupled with the Radiative Beam sub-module. Sub-Kelvin heating of a FLASH-RT dose (approximately 10 mK temperature difference achieved by a 40 Gy/s dose rate beam) is shown by the light yellow-white color, and the temperature’s slow diffusion to the rest of the water phantom can be observed along the beam’s boundaries.

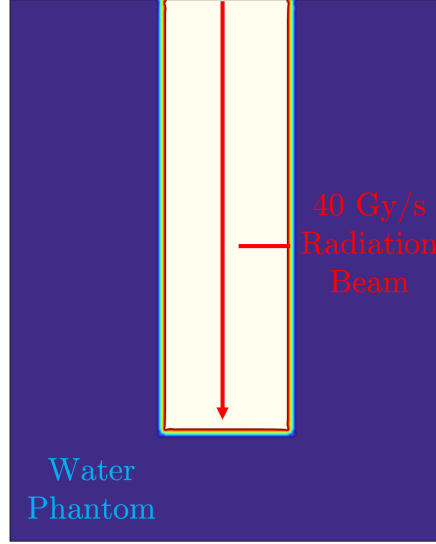


Figure 5: Sub-Kelvin Heat Contours from a FLASH-RT Radiative Beam

Acoustic Model Validation

We can take the water phantom setup shown in Figure 5 and position it on the right of the Acoustic model, shown in Figure 6. This model will simulate an ultrasonic signal, modified from a High-Intensity Focused Ultrasound (HIFU) macro, entering the water phantom from an air ambient outside at a $1 \mu\text{s}$ timepoint, with dimensions modeled after the “fish-tank” bath used in experiment. Simulation of the same ultrasonic signal at $60 \mu\text{s}$ is shown in Figure 7, where localization of normalized Acoustic Pressure in the less-dense heated region can be observed. Thus we see that the Acoustic Pressure plot behaves as expected.

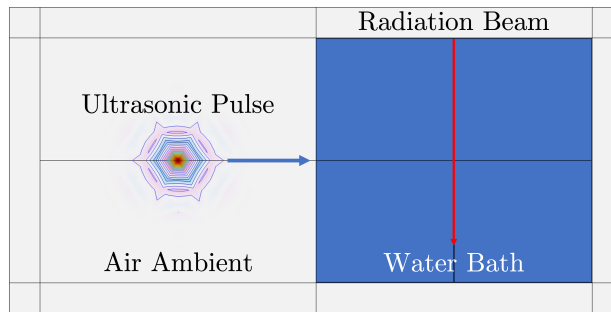


Figure 6: Ultrasonic Signal at $1 \mu\text{s}$

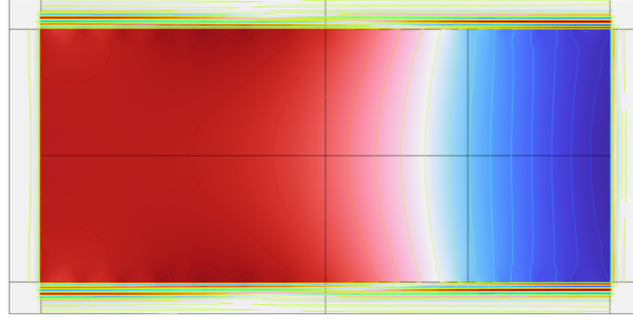


Figure 7: Ultrasonic Signal at $60 \mu s$

Complete Thermoacoustic Model

We can now plot Normalized Acoustic Pressure coupled with Heat Transfer, where an increasingly-red color hue indicates a normalized value close to +1 (a peak), and an increasingly-blue color hue indicates a normalized value close to -1 (a trough). Total Acoustic Pressure is analogous to a change in phase, since a trough in the wave is analog to a compression (90° phase), and a peak in the wave is analog to a rarefaction (270° phase). Furthermore, we can simultaneously analyze a cross-sectional slice of our water phantom apparatus used in experiment – the Surface Plot – with a cut-line of the ultrasonic signal propagated – the Line Plot. We can now proceed to analyze signal timepoints 1, 4, 47, and $60 \mu s$ after propagation, respectively.

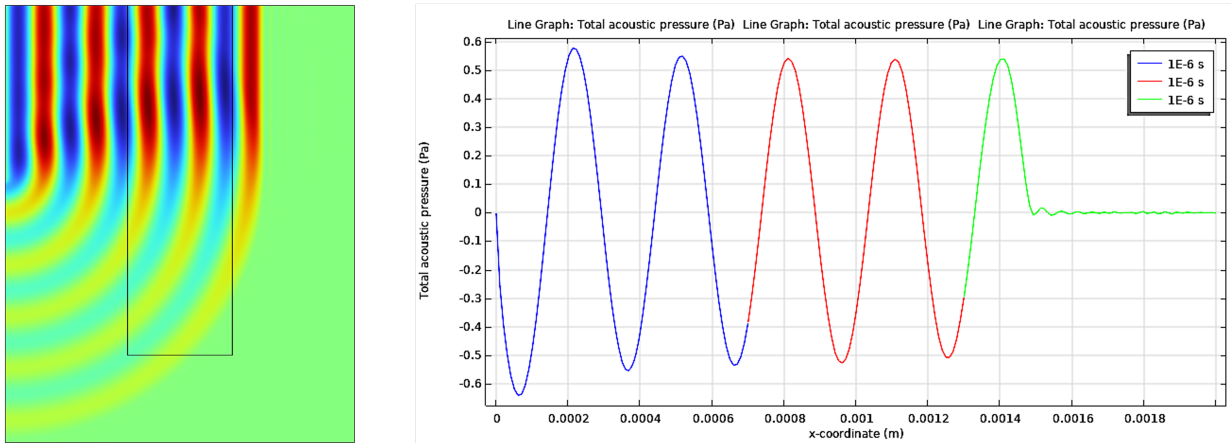


Figure 8: Surface and Line Plot of Total Acoustic Pressure at $1 \mu s$

The Surface Plot on the left of Figure 8 shows a 2-dimensional cross-section of the 3-dimensional water phantom, with the 5 MHz Ultrasonic carrier wave passing through the FLASH-RT irradiated region (boxed, to represent the electron-beam irradiating a tumor cell at the bottom of the water

phantom). Further reducing the dimension by taking a cut-line of the Ultrasonic signal yields the Line Plot on the right, where the blue segment is the signal pre-irradiation, the red segment is the signal during irradiation, and the green segment is the signal post-irradiation. Since no signal has been propagated during $0 \mu s$, our starting timepoint for the signal is $1 \mu s$, as shown above.

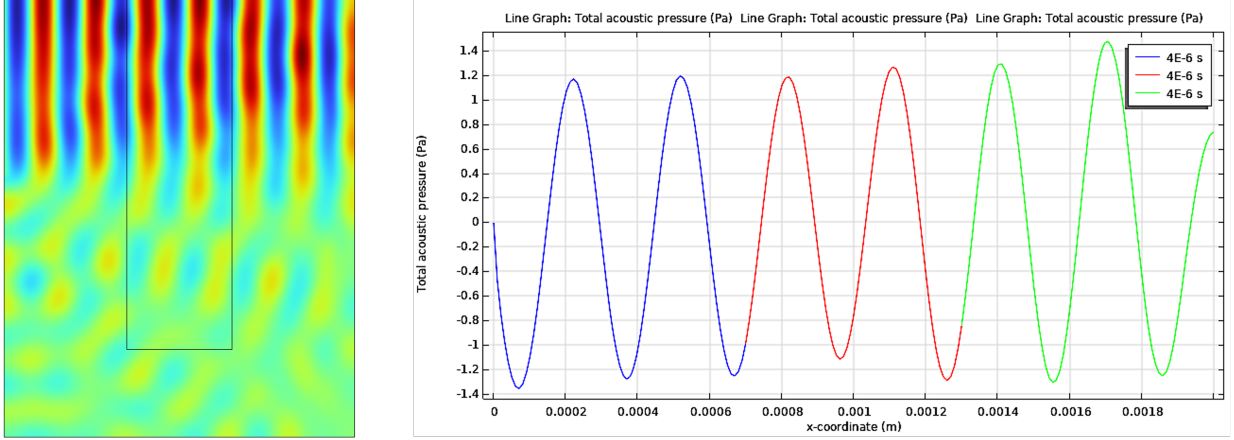


Figure 9: [Control Group] Surface and Line Plot of Total Acoustic Pressure at $4 \mu s$ of Propagation

At $4 \mu s$, we start to see the ultrasonic signal completely cross both ends of the water phantom. At this timepoint, the ultrasonic carrier appears relatively uniform as seen in the Line Plot (right of Figure 9), without minimal disruptions from irradiation. Hence we can call the above Line Plot of Total Acoustic Pressure at $4 \mu s$ after propagation our Control Group: it will serve as our standard to measure effects of phase modulation at future timepoints.

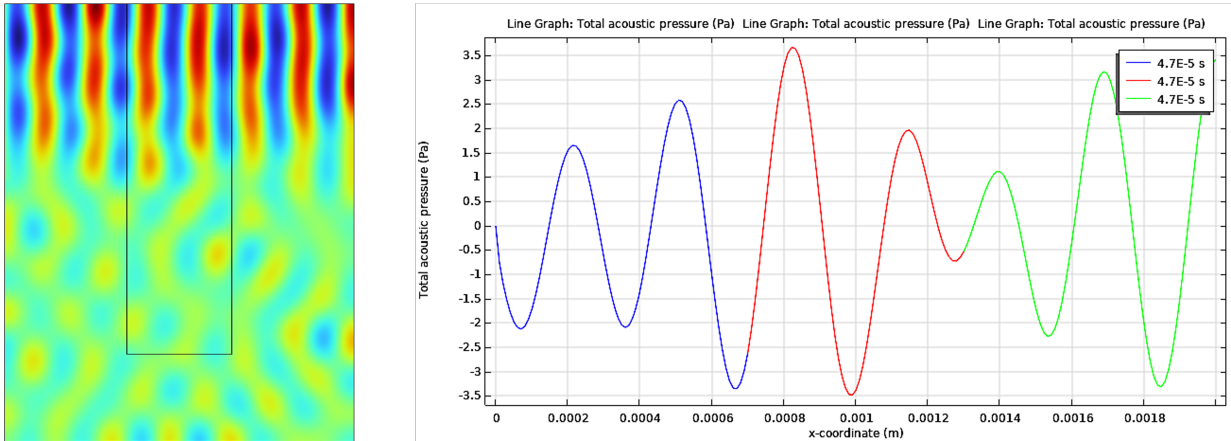


Figure 10: Surface and Line Plot of Total Acoustic Pressure at $47 \mu s$

At an arbitrary timepoint of $47 \mu\text{s}$ after propagations, in Figure 10, we start to see effects of distortion in the original ultrasonic carrier wave. From the Surface Plot (left), it can be observed that the region around and below the FLASH-RT beam starts to resemble an interference pattern. From the Line Plot (right), we begin to see phase modulation effects, which are further amplified at $60 \mu\text{s}$ after propagation:

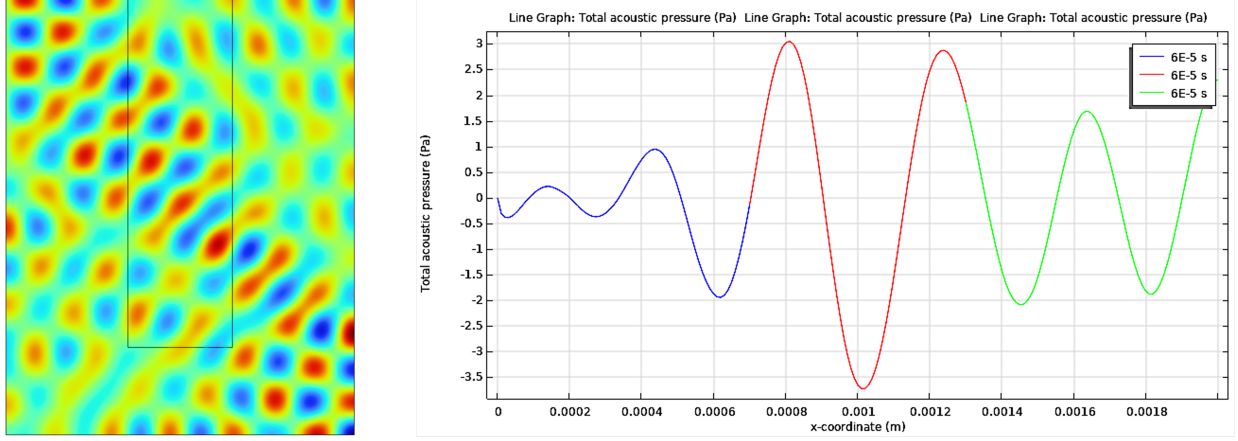


Figure 11: Surface and Line Plot of Total Acoustic Pressure at $60 \mu\text{s}$

From the Surface Plot (left of Figure 11), we see a relatively-large degree of distortion in the carrier wave, with the wave seemingly-compressed in various regions and stretched in others. This observation is consistent in the Line Plot (right), and can be clearly seen upon comparing the above $60 \mu\text{s}$ after propagation Line Plot of Total Acoustic Pressure with our Control Group ($4 \mu\text{s}$ after propagation):

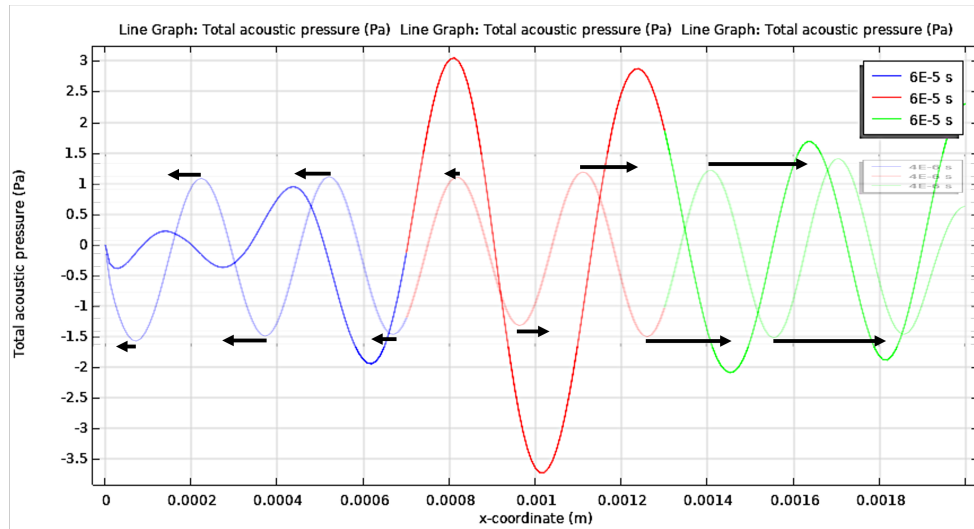


Figure 12: Comparison between Total Acoustic Pressure at $4 \mu\text{s}$ [Control Group] and $60 \mu\text{s}$

As seen above in Figure 12, the ultrasonic signal experiences compression in the blue segment (*i.e.* pre-irradiation), and experiences rarefaction starting in the red segment and primarily in the green segment (*i.e.* during irradiation and post-irradiation). Momentarily treating these two timepoints as two different signals, we can compute the phase dilation factor between the signals as the ultrasonic carrier wave exits the irradiated region to be approximately 2.9; theoretically computing the same measure yields a dilation of 3 upon a $\Delta\phi \approx 18 \frac{\text{mdeg}}{\text{Gy}} \cdot 40 \text{ Gy} = 0.72^\circ$ change in phase.

Therefore, significant phase modulation effects were observed through both the theoretical and computational model frameworks. Additionally, amplitude modulation effects were also observed (primarily in the irradiation zone), which is an effect in close agreement with preliminary results observed in experiment. Hence, the theoretical framework produced justifies the use of our novel detection principle’s hardware (ultrasonic imaging via piezoelectric transducers) to measure phase modulation instead of Time of Flight, and suggests an 18 millidegree change in phase per Gray. The computational framework produced validates this expected phase modulation, allowing for us to expect phase modulation effects in reality. As such, oncologists, specifically FLASH-RT practitioners, only need to add a set of piezoelectric transducers on either side of the patient’s targeted tumor-region and connect the transducers to a phase-demodulation instrument (relatively-low-grade lock-in amplifier, radio frequency spectrum analyzer, etc.) to determine, with precision, the radiation dose and dose-rate received by the tumor cells. A proposed experimental setup is detailed below in Figure 13, with the water phantom again replacing tumor cells, and has been used in preliminary experiments at the Medical Industrial Radiation Facility (MIRF) at the National Institute of Standards and Technology (NIST):

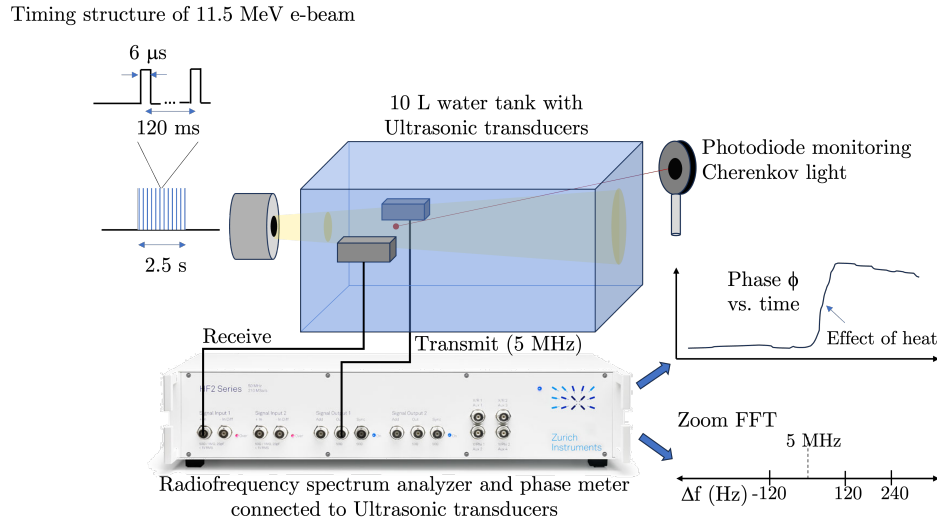


Figure 13: Experimental Setup

“Ultrasonic Calorimetry for Dose to Water in FLASH Electron Beams, Figure 1.” Graphic created by Dr. Ronald E. Tosh (Research Advisor), National Institute of Standards and Technology, 2024.

Discussion and Conclusions

Discussion

The findings of this research study are based in an idealized environment, created via the COMSOL® Multiphysics 6.1 Software. While this enables results to appear almost hyper-accurate, it introduces bias when extrapolating the results discussed to realistic setups. Thus, the results indicated in this report, while promising in its ability to be closely replicated in experiment, are prescriptions for experimental results.

However, the design modeled has great versatility for application in experimental setups. Since the phase dilation factor determined in simulation (2.9) was in close agreement with the theoretical factor of 3, the design created hence has great practicality and can be used as an accurate dose and dose-rate detection principle for Ultra-High Dose Rate applications, or more broadly, as an Absolute Dosimetry standard.

During the testing of simulations on the COMSOL® Multiphysics 6.1 software, the computational power and memory space required for thorough functionality exceeded the limits of the device used. Hence, the MUMPS-out-of-core module was used for extra memory. It is unknown whether this action introduced bias in data filtration, but more-conclusive simulations would require the use of a supercomputing device for its computational power and memory.

Thermoacoustic shocks were observable in models that had the presence of a cancerous tissue cell (not pictured in this report). It is unknown if this is an effect of the fine meshing used by the Finite-Element Method (FEM) scheme; regardless, this effect agrees with behaviours reported in literature [21, 22].

Similar to an experiment, repeated testing of the same simulation would yield similar values in the Normalized Acoustic Pressure Plots; occasionally, the plots would return with significantly varied results, which were deemed as erroneous plots resulting from error present in the study. Simulated results of this research study agreed with preliminary experimental testing, and hence are expected to scale to a real-world experimental setup.

Conclusion

The dose and dose-rate detection principle was simulated via a complete Thermoacoustic model on the COMSOL® Multiphysics 6.1 software, modeled after equipment readily available at the

Medical Industrial Research Facility (MIRF). By propagating a secondary type of radiation, Ultrasound, as a diagnostic modality orthogonal to the therapeutic modality FLASH-RT electron beam, significant effects of phase modulation were observed, enabling the imaging of the radiation dose delivered to the patient. Hence, the detection scheme developed has the potential to be used in an experimental setup to precisely measure FLASH-RT doses with minimal additional equipment (including two piezoelectric transducers and a lock-in amplifier), allowing for this model to have versatility in physical applications. Thus, the detection principle developed not only is efficient in precisely measuring Ultra-High Dose Rate (UHDR) dosimetry in simulations, but also has the potential to be efficient in clinical trials of FLASH-RT and related phenomena.

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